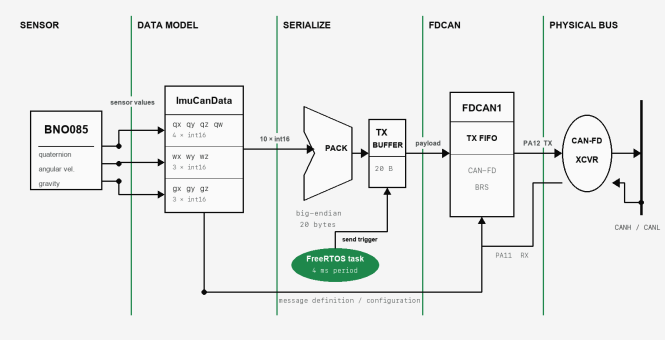
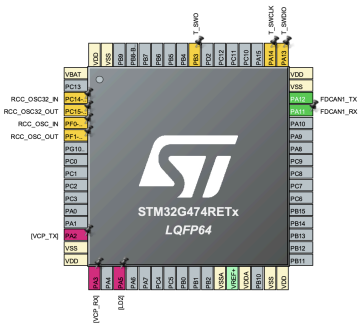
Portfolio

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Humanoid IMU to CAN FD Firmware – WATonomous



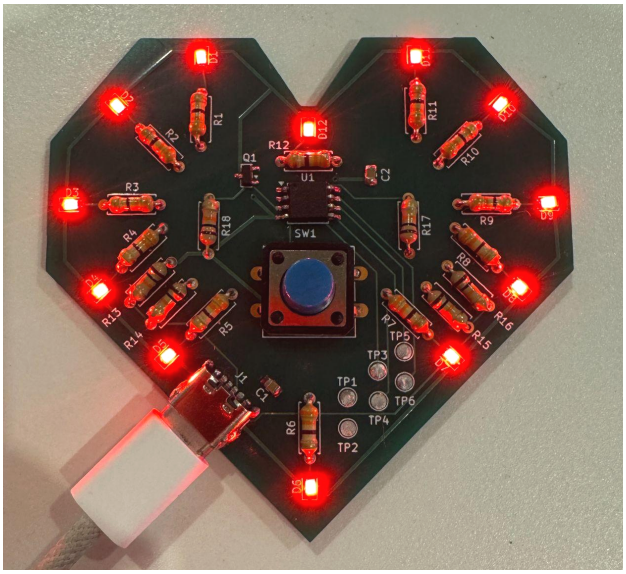
IMU-to-CAN-FD Firmware Data Path



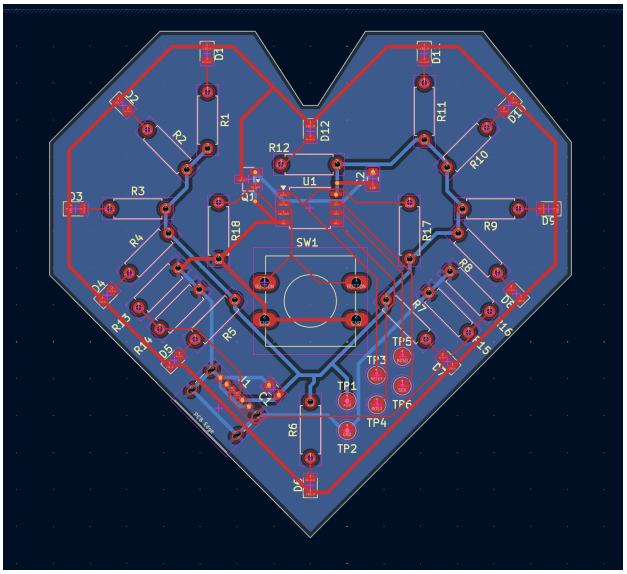
STM32G474RE FDCAN Pin Configuration

- Defined a 20-byte CAN-FD payload representing quaternion, angular-velocity, and gravity measurements as ten signed 16-bit signals from a BNO085 IMU
- Implemented STM32G474 FDCAN1 initialization and transmission logic, including big-endian serialization, a 20-byte transmit buffer, and CAN-FD bit-rate switching
- Created a FreeRTOS publishing task using vTaskDelayUntil() with a configured 4 ms transmission period
- Added matching DBC message and signal definitions for decoding the IMU payload within the humanoid robotics software stack
- Technical documentation and code available here

Programmable Heart LED PCB



Assembled PCB



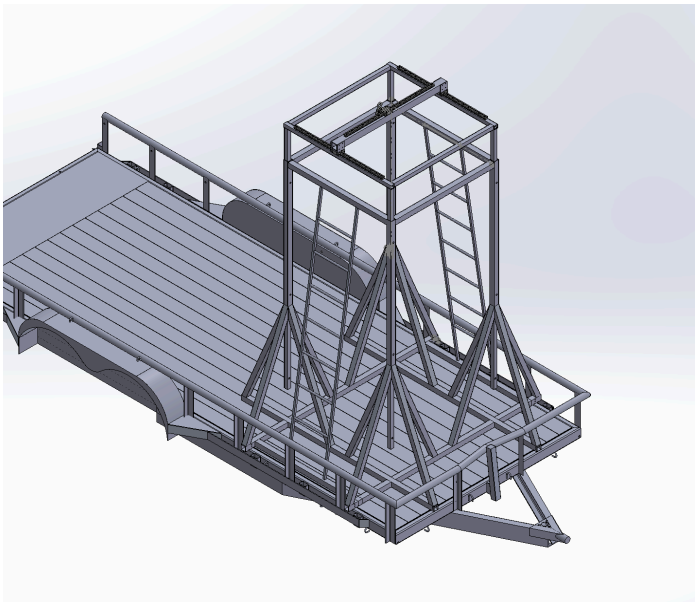
KiCAD PCB Layout

- Designed and assembled a custom 12-LED heart-shaped PCB in KiCad, integrating 5 V USB-C power, an ATtiny85, and MOSFET-switched PWM control
- Programmed and debugged embedded C++ firmware to illuminate 12 LEDs at power-on and execute a button-triggered PWM sequence with debounced input, timed pulses, and rapid fade-to-dim transitions
- Hand-assembled and debugged the PCB, isolating solder bridges and connection faults through continuity testing and reworking the board with flux and solder wick
- Integrated current-limiting resistors, decoupling capacitors, pull-up and pull-down resistors, programming connections, and MOSFET switching into a functioning microcontroller-based system
- Technical documentation and code available here

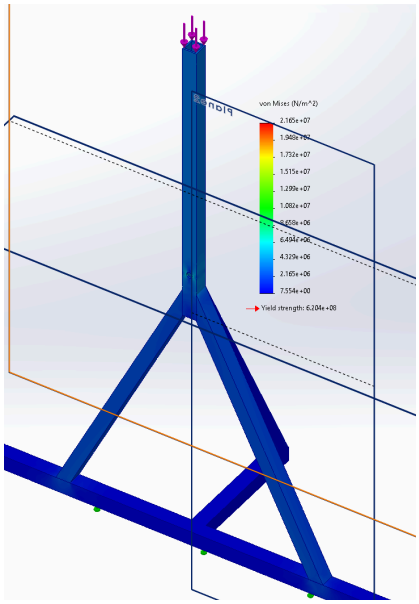
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Trailer-Mounted Oxidizer Tank Support Tower – Waterloo Rocketry



CAD Rendering of Tank Tower Mounted on Trailer



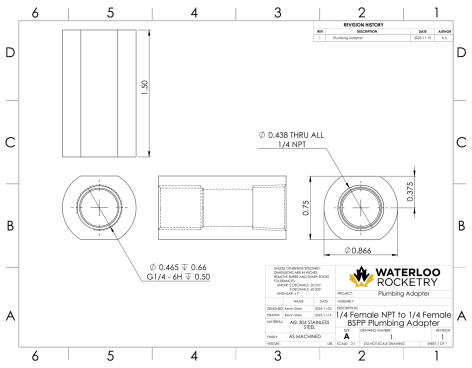
Stress Test Load Simulation

- Designed and modeled a preliminary steel oxidizer tank tower for Waterloo Rocketry, developing the concept through PDR with an approved project budget of $5000
- Defined structural requirements for a 500 kg static load capacity, providing a factor of safety greater than 6 for an 80 kg propellant load while maintaining a freely suspended tank for accurate load-cell measurements
- Developed a telescoping gantry architecture with lockable X, Y, and Z adjustment, combining structural steel tubing, guide-rail carriages, and a hoist system to align the oxidizer tank with feed-system plumbing
- Evaluated critical structural load cases including beam deflection, weld loading, and trailer-mounting loads, performed FMEA and proposed max-load, motion, and vibration validation testing
- Preliminary Design Review presentation available here

Plumbing Adapter – Waterloo Rocketry



Machined Adapter with Thread



Technical CAD Drawing

- Designed a 1/4” BSPP female to 1/4” NPT female stainless steel adapter for fuel fluid systems; contributed to world record amateur liquid rocket
- Used SolidWorks to create proper drawings and tolerances, then machined the part on lathe and mill

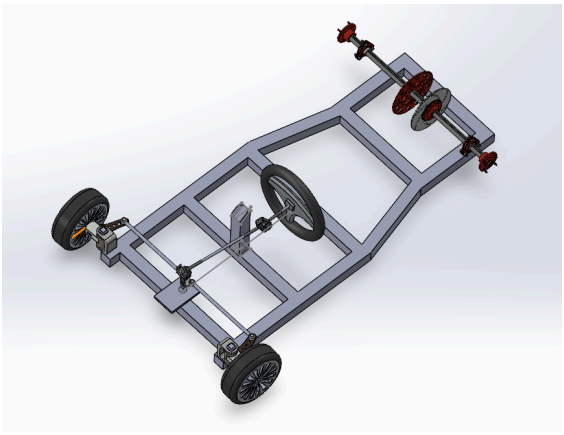
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Electric Go-Kart Chassis Design & Fabrication



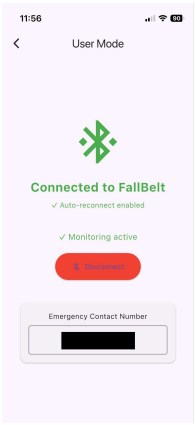
Fabricated Chassis with Installed Steering System



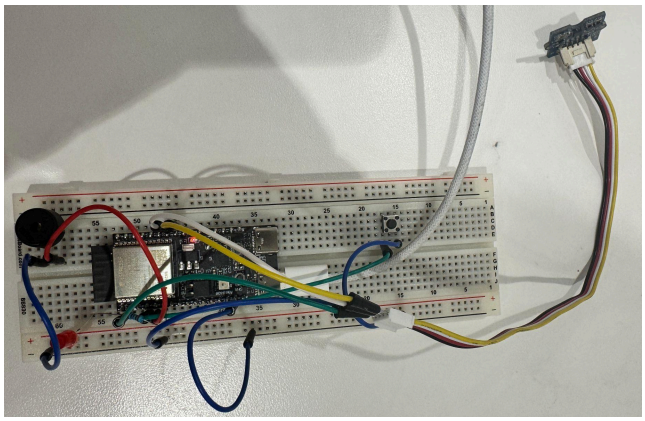
SolidWorks Chassis & Component Assembly

- Designed the go-kart chassis in SolidWorks and built the full-scale frame from steel tubing and welded the structure from the CAD design
- Mounted and connected the rack-and-pinion steering system, including the steering wheel, segmented column, universal joints, adjustable tie rods, and front spindles
- Next steps include adding rear axle and wheels, implementing the drivetrain, and developing autonomous driving

Wearable Fall Detection & Emergency Response System Prototype – DeltaHacks 12



Flutter Mobile App Connected via BLE



ESP32 Microcontroller Fall Detection Prototype

- Co-developed a wearable fall-detection system integrating an ESP32-C6, 3-axis accelerometer, Bluetooth Low Energy, and Flutter mobile application
- Developed the Flutter/Dart BLE interface to discover and connect to the FallBelt, subscribe to GATT notifications, and display real-time fall alerts
- Implemented a multi-stage emergency response flow allowing users to cancel false alarms, request assistance, or escalate unanswered alerts
- Integrated GPS, Firebase, and Google Maps to share active help requests with nearby volunteers
- Devpost available here and code available here

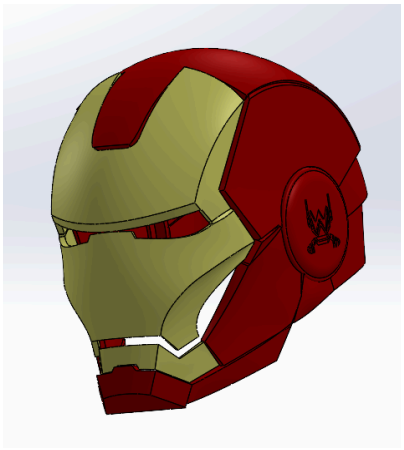
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Iron Man Helmet Prototype - WATBotics



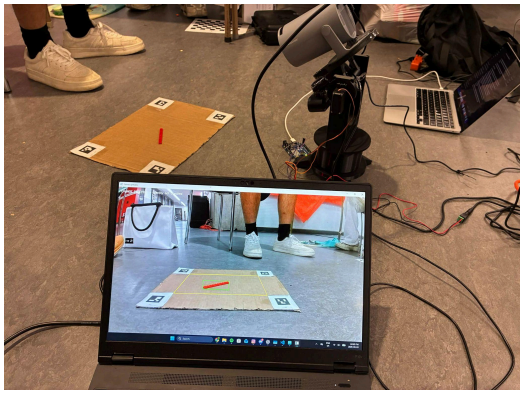
3D Printed and Assembled Helmet



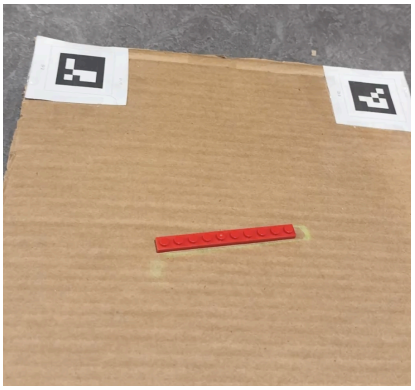
CAD Rendering of Helmet Assembly

- Co-developed a full-scale Iron Man helmet prototype in SolidWorks with a 3D-printed internal frame and removable cosmetic shell
- Led the internal packaging design by defining component locations, wiring paths, mounting space, and hardware clearances within the helmet
- Evaluated 5 candidate polymers for the frame, shell, and visor based on toughness, heat resistance, surface finish, and printability
- Contributed to 3D printing and final assembly, checking the fit between the structural frame, cosmetic shell, and internal hardware

Projector-Guided Assembly System – Hack the North 2026



Assembled Camera/Projector Mechanism with Live Object



Projected Highlight on Assembly Workspace

- Co-developed a CAD-guided assembly assistant combining OpenCV placement verification with projected cues that identify which component to select and where to place it
- Designed and built a two-axis camera/projector mechanism with 3D-printed servo couplings, bearing supports, and bolted interfaces, integrating a teammate's mounting plate and tilt shaft
- Programmed Arduino C++ firmware to control two MG996R servos through a PCA9685 over I2C, with independent PWM limits and home positions
- Implemented USB serial commands for axis positioning, status reporting, and output disable to support hardware setup and integration
- Devpost available here and code available here